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Real Effects of Centralized Markets: Evidence from Steel Futures

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1 Big picture

- **Question.** Do centralized derivative markets affect the underlying product markets, not just the financial markets in which derivatives trade?
- **Empirical setting.** The paper studies the staggered introduction of NYMEX futures contracts for two U.S. steel products: hot-rolled coil (HRC) futures in 2008 and busheling scrap (BUS) futures in 2012.
- **Core economic mechanism.** Centralized futures markets can affect physical product markets through two linked channels: a **reference price channel**, because futures prices create public benchmark prices for negotiations, and a **risk-management channel**, because exchange-traded futures may make hedging easier.
- **Reference price predictions.** If buyers can use futures prices as reference prices in bilateral negotiations, price dispersion should fall, low-cost producers should gain market share, and product prices should decline.
- **Risk-management predictions.** If futures improve hedging, producers can stabilize cash flows and compete more aggressively for market share. If futures instead destroy risk-sharing opportunities by revealing information too broadly, hedging could worsen. The sign is therefore empirical.
- **Main identification strategy.** The paper uses difference-in-differences comparisons between steel products that receive futures contracts and related steel products that were also plausible candidates for futures trading but did not receive a contract during the sample.
- **Main result 1: price transparency.** Price dispersion in the physical steel market falls sharply after futures introductions: the coefficient of variation declines by about 5.6 to 5.9 percentage points.
- **Main result 2: risk management.** Firms selling treated products become more likely to discuss commodity derivatives in annual reports, suggesting that centralized futures increase producer hedging.
- **Main result 3: product market competition.** Market share becomes more sensitive to relative input costs, consistent with low-cost producers gaining share after buyers obtain better reference prices.
- **Main result 4: prices and producers.** Product prices fall by roughly 3–4 percent, operating profits decline by about 1.6–2.2 percentage points of assets, and affected producers experience negative abnormal stock returns around futures-related news.
- **Takeaway.** Centralized futures markets improve transparency and hedging, but in this setting those benefits intensify product market competition and reduce producer rents.

2 Data and measurement

2.1 Futures introductions and treatment definition

- The treated product in the first event is **hot-rolled coil (HRC)**, with NYMEX/CME futures trading beginning in 2008.
- The treated product in the second event is **busheling scrap (BUS)**, with NYMEX/CME futures trading beginning in 2012.
- Product-level treatment is captured by $\text{FuturesProduct}_{k,p}$, equal to one for HRC in the HRC event window and BUS in the BUS event window.
- Firm-level treatment is captured by $\text{FuturesFirm}_{k,i}$, equal to one for firms selling HRC around the HRC introduction or firms selling ferrous scrap around the BUS introduction.
- The control products are related steel products tracked by SteelBenchmarker and initially plausible candidates for U.S. futures trading: cold-rolled coil, plate, heavy melting scrap, and shredded scrap.

Interpretation: The key idea is not random assignment of futures contracts. Instead, treatment is plausibly comparable because both treated and control products were candidates for futures trading and were exposed to common steel-market shocks.

2.2 Product-level price data from SteelBenchmarker

- The product-level data come from SteelBenchmarker, a price reporting service created to support reliable steel price indices.
- The data cover six steel products sold to U.S. customers: HRC, cold-rolled coil, standard plate, busheling scrap, heavy melting scrap, and shredded scrap.
- Hot-rolled and cold-rolled coil data begin in the first week of 2007; standard plate data begin in 2008; scrap product data begin in 2009.
- Price indices are published twice per month, on the second and fourth Wednesdays.
- Reporting firms submit transaction prices over a five-day window before publication.
- The key price outcomes are the price level, log price, the cross-reporter standard deviation of reported prices, and the coefficient of variation of reported prices.

Interpretation: The data are particularly valuable because prices in physical steel markets are negotiated bilaterally rather than posted on a centralized spot market. Cross-reporter dispersion is therefore a direct measure of opaque bargaining outcomes.

2.3 Firm-level annual report, accounting, and stock data

- Firm product exposure is manually identified from annual reports.
- For the HRC event, treated firms are those that report selling hot-rolled coil before HRC futures trading starts.
- For the BUS event, treated firms are those that report operating a ferrous scrap business before BUS futures trading starts.
- The paper manually checks that firms are sellers rather than buyers of the treated products and restricts the sample to metal-producing and recycling industries.
- Hedging is measured by whether the annual report discusses commodity derivative use.

- Firm accounting outcomes come from quarterly Compustat data.
- Stock price reactions are measured using CRSP daily stock returns around specific news events that increase the likelihood of steel futures introductions.

Interpretation: The paper combines product-level evidence on prices with firm-level evidence on hedging, market share, profits, and valuations. This is important because the mechanisms operate at different levels: buyers use prices in product negotiations, while producers respond through hedging and competitive behavior.

2.4 Price dispersion outcomes

For product p and publication date t , the paper computes price dispersion across price reporters:

$$SD(Price)_{p,t} = \text{sd}\{Price_{r,p,t} : r \in \mathcal{R}_{p,t}\},$$

$$CV(Price)_{p,t} = \frac{SD(Price)_{p,t}}{Price_{p,t}}.$$

Interpretation: The standard deviation captures absolute dispersion in dollars per ton. The coefficient of variation scales dispersion by the average price, making it comparable across products and over periods with different steel price levels.

2.5 Risk management and hedging variable

- $\text{Hedge}_{i,y}$ is an indicator equal to one if a firm discusses using commodity derivatives in its annual report.
- The variable captures reported hedging activity, not the exact volume or delta of hedges.
- The annual report measure is useful because many industrial firms do not disclose standardized futures positions in a way that is comparable across firms.

Interpretation: The hedging variable should be interpreted as evidence that futures availability makes commodity derivative usage salient or feasible. It is not a perfect measure of hedge intensity, but the timing and treatment-control comparison make it informative.

2.6 Market share and cost advantage variables

- The paper studies market share within four-digit NAICS industries.
- Steel producers differ in production technology: basic oxygen furnace (BOF) producers primarily use iron ore, while electric arc furnace (EAF) producers primarily use scrap.
- The quarterly ratio of iron ore prices to scrap prices measures the time-varying relative cost advantage of EAF versus BOF producers.
- The market-share test focuses on HRC, because the contrast between EAF and BOF technology is central to measuring relative cost advantage.

Interpretation: This design translates the idea that buyers can better identify low-cost sellers into a measurable prediction: when EAF technology becomes relatively cheaper, EAF producers should gain more market share after the arrival of futures prices.

2.7 Profit and valuation outcomes

- Operating profit is measured as operating income before depreciation scaled by beginning-of-quarter total assets.
- The event-study outcome is cumulative abnormal return over a five-day event window around news about futures introductions.
- CARs are computed using several benchmarks: market-adjusted returns, CAPM, Fama-French three-factor, and Carhart four-factor models.

Interpretation: The profit and valuation tests ask whether greater transparency and hedging ultimately benefit producers or instead erode producer rents through stronger product market competition.

3 Models and empirical specifications

3.1 Product-level price dispersion DiD

The main price-dispersion specification is:

$$\text{PriceDispersion}_{k,p,t} = \beta \text{Post}_t \times \text{FuturesProduct}_{k,p} + \alpha_{k,p} + \alpha_{k,t} + \varepsilon_{k,p,t}. \quad (1)$$

- k indexes the futures introduction: HRC or BUS.
- p indexes product.
- t indexes SteelBenchmarker publication dates.
- $\alpha_{k,p}$ are futures-introduction-specific product fixed effects.
- $\alpha_{k,t}$ are futures-introduction-specific publication-date fixed effects.
- β measures how price dispersion changes for treated products after the futures introduction relative to control products.

Interpretation: This specification compares treated and control products within the same event window and publication dates. Date fixed effects absorb shocks common to all products, such as overall steel demand or commodity price cycles.

3.2 Dynamic price-dispersion event study

$$CV(\text{Price})_{k,p,t} = \sum_{\tau=-5}^5 \beta_{\tau} \text{FuturesProduct}_{k,p} \mathbf{1}\{t = \tau\} + \alpha_{k,p} + \alpha_{k,t} + \varepsilon_{k,p,t}, \quad (2)$$

where $t = -1$ is omitted as the reference period and endpoints combine $t \leq -5$ and $t \geq 5$.

Interpretation: The event study checks whether the post-treatment decline in price dispersion is preceded by pre-trends. Parallel pre-trends strengthen the interpretation that futures introductions, rather than earlier product-specific shocks, drive the results.

3.3 Firm-level hedging DiD

The hedging specification is:

$$\text{Hedge}_{k,i,y} = \beta \text{Post}_y \times \text{FuturesFirm}_{k,i} + \gamma' X_{k,i} \times \tau_y + \alpha_{k,i} + \alpha_{k,j,y} + \varepsilon_{k,i,y}. \quad (3)$$

- i indexes firms, j indexes three-digit NAICS industry, and y indexes years.
- $X_{k,i}$ includes baseline controls such as firm size, age, and sales growth.
- $\alpha_{k,i}$ are futures-introduction-specific firm fixed effects.
- $\alpha_{k,j,y}$ are futures-introduction-specific industry-year fixed effects.

Interpretation: The specification asks whether firms exposed to treated steel products change their hedging disclosures after futures trading becomes available, relative to firms in the same broad industries but without treated-product exposure.

3.4 Market share and cost sensitivity

The market-share specification is:

$$MarketShare_{i,q} = \beta Post_q \times FuturesFirm_i \times EAF_i \times Iron/Scrap_q + \alpha_i + \alpha_q + \alpha_{j,q} + \varepsilon_{i,j,q}. \quad (4)$$

- EAF_i indicates producers using electric arc furnaces, whose main input is scrap.
- $Iron/Scrap_q$ is the quarterly ratio of iron ore prices to scrap prices.
- A positive coefficient means that after futures introduction, EAF producers gain more market share when scrap becomes cheap relative to iron ore.

Interpretation: This is a mechanism test for competition. If futures prices help buyers find low-cost suppliers, market share should become more responsive to relative production costs.

3.5 Product price DiD

$$\ln(Price)_{k,p,t} = \beta Post_t \times FuturesProduct_{k,p} + \alpha_{k,p} + \alpha_{k,t} + \varepsilon_{k,p,t}. \quad (5)$$

Interpretation: Lower product prices after futures introductions are consistent with reduced seller markups and better buyer bargaining. The effect also captures cost reallocation if market share shifts toward lower-cost producers.

3.6 Operating profit DiD

$$Profit_{k,i,q} = \beta Post_q \times FuturesFirm_{k,i} + \gamma' X_{k,i} \times \tau_q + \alpha_{k,i} + \alpha_{k,j,q} + \varepsilon_{k,i,q}. \quad (6)$$

Interpretation: This test asks whether better hedging and information ultimately improve producers' profitability or whether stronger competition compresses margins. Negative profit effects indicate that competitive pressure dominates any production-planning benefits.

3.7 Stock market event study around futures news

The event-window regression is:

$$CAR_{i,e} = \beta FuturesFirm_{i,e} + \alpha_e + \alpha_i + \varepsilon_{i,e}. \quad (7)$$

The dynamic event-study version is:

$$CAR_{e,i,d} = \sum_{\tau=-10}^{10} \beta_{\tau} FuturesFirm_{e,i} \mathbf{1}\{d = \tau\} + \alpha_{e,i} + \alpha_{e,d} + \varepsilon_{e,i,d}. \quad (8)$$

Interpretation: The stock market tests are important because they use short windows around discrete news, limiting confounding by slow-moving product market conditions. Negative CARs suggest investors expect steel futures to reduce producer rents.

4 Identification and empirical design

4.1 Core identification assumption

The main identifying assumption is a conditional parallel-trends assumption: absent the introduction of futures markets, treated steel products and comparable control products would have followed similar trends in price dispersion, prices, and firm outcomes.

Interpretation: The paper does not require futures introductions to be random. It requires that the timing of introductions not coincide with unobserved shocks that differentially affect treated products relative to comparison products in the same direction as the estimated effects.

4.2 Why the control products are credible comparisons

- Control products are related steel products tracked by the same data provider.
- Several control products were initially considered candidates for U.S. futures trading.
- Treated and control products had similar ex ante volatility, a key condition for futures viability.
- Treated and control products are exposed to common steel-market shocks, allowing date fixed effects to absorb aggregate demand, commodity, and macro conditions.

Interpretation: The control group is not arbitrary. It consists of steel products close enough to be affected by the same macro and industry forces, but without the same introduction of centralized futures during the focal event window.

4.3 Evidence supporting parallel trends

- Event-study plots show no clear pre-trends in price dispersion, hedging, product prices, profits, or valuations before the futures introductions.
- Treated and control products are similar in cyclical and volatility before treatment.
- Non-U.S. producers selling the same treated products do not show similar post-introduction changes, suggesting the effects are specific to the U.S. futures market rather than global product shocks.

Interpretation: The combination of pre-trend evidence, ex ante comparability, and geographic placebo tests is central to the causal interpretation.

4.4 Mechanism identification

- The price-dispersion results identify the **reference price channel**: if futures prices improve transparency, cross-reporter price dispersion should fall.
- The hedging results identify the **risk-management channel**: if centralized futures improve hedging opportunities, treated producers should increase commodity derivative usage.
- The market-share-cost sensitivity test connects these channels to **competition**: improved information and hedging should allow low-cost producers to gain share.
- The price, profit, and valuation tests then assess the net effect on the product market and producers.

Interpretation: The empirical design is strongest because different outcomes line up with a single mechanism: transparency and hedging increase competition, which lowers prices and producer rents.

4.5 Confounding-event strategy

The paper addresses possible confounds from the Great Recession, commodity cycles, input prices, import competition, mergers, and product-specific shocks using several tests:

- demand, supply, and trade controls in product-level regressions;
- separate estimation for HRC and BUS, so the BUS results avoid the 2008 crisis window;
- tests of production quantities to check whether lower prices reflect demand contractions;
- controls for input price exposure and import competition;
- exclusion of firms involved in major acquisitions;
- non-U.S. placebo tests using firms exposed to similar global steel conditions but not the U.S. NYMEX futures market.

Interpretation: The identifying challenge is that HRC futures began during the financial crisis. The BUS-only results and non-U.S. placebo tests are therefore especially important because they show similar mechanisms outside the most concerning confounding window.

4.6 Spillover concern

- Control products may also benefit indirectly from reference prices or better hedging if buyers and producers use futures prices as broader steel benchmarks.
- Such spillovers would likely move control products in the same direction as treated products.
- Therefore, spillovers would attenuate the estimates rather than mechanically generate the observed treated-control differences.

Interpretation: The results are best interpreted as effects on directly treated products relative to related products. If indirect effects also reach controls, the paper’s estimates may be lower bounds.

4.7 Limits of the design

- Futures introductions are not randomized.
- There are only two main treated product introductions, so inference must address clustered and aggregate shock concerns carefully.
- Hedging is measured from annual-report disclosure rather than exact hedge positions.
- Product market outcomes are observed for selected steel products and may not generalize to all derivative markets.
- The paper observes reduced producer profits but cannot fully decompose all welfare gains between buyers, low-cost firms, and end consumers.

Interpretation: The paper provides credible evidence of real effects from centralized futures markets, but it is not a randomized experiment and should be read as evidence from a specific but economically important market setting.

5 Tables and figures

5.1 Figure 1: Growth of Exchange Trading of Derivatives Over Time and Table 1: Summary Statistics

Read:

- Figure 1 shows the long-run growth of exchange-traded derivatives and commodity futures, motivating the question of whether centralization affects real product markets.
- Table 1 shows that 32.3% of product-level observations are treated and 4.0% of firm-level observations are treated.
- The mean steel price is about \$710 per ton, while average price dispersion is large: $SD(Price) = 125.8$ and $CV(Price) = 0.187$.
- The average firm in the pre-period has assets of about \$2.4 billion, age of 38 years, and sales growth of 2.2%.
- Event-study CARs are negative on average, foreshadowing the later valuation results.

Interpretation: The summary statistics show that physical steel markets are both economically large and opaque. The baseline price dispersion is high enough for a reference-price mechanism to matter, and the firm sample includes sizable industrial firms for which changes in hedging and competition can be economically meaningful.

Economic message: The paper's empirical setting is well suited to studying real effects of market centralization: prices are negotiated, transparency is limited, and futures prices can plausibly become benchmarks.

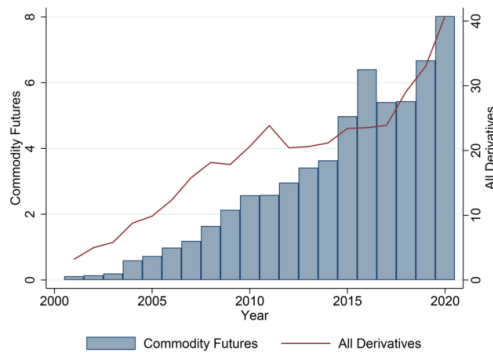


Figure 1
Growth of Exchange Trading of Derivatives Over Time
This figure shows the evolution of global exchange trading in derivatives (in billion contracts) for all derivatives (red line, right axis) and for commodity futures (blue bars, left axis). *Source:* World Federation of Exchange

In the empirical tests, I leverage information on firms' production technology combined with fluctuating prices of iron ore versus scrap, to measure firm

(a) Figure 1: Exchange-traded derivatives and commodity futures growth

Table 1
Summary statistics

Variable:	(1) Obs.	(2) Mean	(3) SD	(4) $p1$	(5) $p50$	(6) $p99$
<i>A. Product-level treatment and outcomes</i>						
Futures _{product}	19,653	0.323	0.468	0.000	0.000	1.000
Price	19,653	709.814	224.202	339.480	705.000	1,315.000
Ln(Price)	19,653	6.514	0.324	5.827	6.558	7.182
SD(Price)	708	125.759	78.921	15.846	120.275	377.361
CV(Price)	708	0.187	0.108	0.038	0.171	0.515
<i>B. Firm-level treatment and outcomes</i>						
Futures _{firm}	2,993	0.040	0.196	0.000	0.000	1.000
Hedge (1/0)	2,993	0.261	0.439	0.000	0.000	1.000
Market share	1,419	0.085	0.136	0.001	0.031	0.717
Profit/Assets	5,095	0.028	0.025	-0.052	0.028	0.113
<i>C. Firm-level pre-period characteristics</i>						
Assets (mn)	355	2.395	5.039	10	755	29,457
Firm age	355	38.020	21.606	1.000	36.000	80.000
Sales growth	355	0.022	0.158	-0.428	0.005	0.585
<i>D. Event study</i>						
Futures _{firm}	1,106	0.040	0.196	0.000	0.000	1.000
CAR _{market-adj.}	1,106	-0.011	0.064	-0.215	-0.004	0.186
CAR _{CAPM}	1,106	-0.014	0.065	-0.225	-0.007	0.182
CAR _{3-factor}	1,106	-0.013	0.066	-0.228	-0.007	0.168
CAR _{4-factor}	1,106	-0.014	0.065	-0.222	-0.009	0.169

This table presents summary statistics for the variables used in the analysis. [Internet Appendix E](#) provides definitions.

represent futures introduction-specific product and publication date fixed

(b) Table 1: Summary statistics

5.2 Table 2: Physical Product Market Price Dispersion and Figure 2: Physical Product Market Price Dispersion - Dynamic Effects

Read:

- Table 2 reports large declines in absolute and relative price dispersion after futures introductions.
- In Panel A, $SD(Price)$ falls by about \$30 to \$39 per ton across specifications.
- In Panel B, $CV(Price)$ falls by about 5.6 to 5.9 percentage points.
- Results survive demand controls, supply controls, trade controls, and all controls together.
- Figure 2 shows that pre-treatment coefficients are close to zero and the decline occurs after the futures introduction.

Interpretation: This is the direct test of the reference price channel. The decline in price dispersion means that bilateral steel prices converge after the arrival of public futures prices. The result is not just a level effect: the dynamic pattern supports the view that the futures market changes bargaining information after introduction.

Economic message: Better public price signals reduce informational frictions in decentralized product markets. Buyers can bargain using a common reference price, limiting sellers' ability to charge very different prices to different customers.

Table 2
Physical product market price dispersion: DiD estimation around futures introductions

	(1)	(2)	(3)	(4)	(5)
A. Standard deviation					
	SD(Price)				
Post×Futures _{product}	−38.636*** (7.737)	−34.695*** (8.835)	−32.220*** (9.362)	−35.349*** (7.946)	−30.339*** (9.232)
R ²	.769	.773	.770	.770	.775
Observations	708	708	708	708	708
B. Coefficient of variation					
	CV(Price)				
Post×Futures _{product}	−0.057*** (0.011)	−0.057*** (0.014)	−0.059*** (0.014)	−0.058*** (0.012)	−0.056*** (0.016)
R ²	.676	.677	.676	.676	.678
Observations	708	708	708	708	708
Product FE	Yes	Yes	Yes	Yes	Yes
Publication date FE	Yes	Yes	Yes	Yes	Yes
Demand controls	No	Yes	No	No	Yes
Supply controls	No	No	Yes	No	Yes
Trade controls	No	No	No	Yes	Yes

This table presents estimates of the effect of centralized futures markets on physical product market price dispersion obtained from a difference-in-differences analysis around futures introductions. Price dispersion is measured for each publication date and product across different reporting firms using the standard deviation ($SD(Price)$) of price submissions in panel A and the coefficient of variation ($CV(Price)$) in panel B. $Futures_{product}$ is an indicator equal to one for hot-rolled coils (HRC) around the 2008 introduction of HRC futures and for busheling scrap (BUS) around the 2012 introduction of BUS futures. Control products are cold-rolled coil, plate, heavy-melting scrap, and shredded scrap. Prices are collected and published by *SteelBenchmark* every 2 weeks. *Post* is an indicator equal to one for publication dates $t=0, 1, \dots, 40$, where $t=-1$ denotes the last publication date before the futures started trading. Column 2 adds controls to capture steel demand: quarterly GDP growth and output growth of key steel-consuming sectors (i.e., automotive, construction, machinery, appliances), interacted with the $Futures_{product}$ indicator (Source: Bureau for Economic Analysis). Column 3 adds controls to capture steel supply: annual growth rates of U.S. steel production (in metric tons) and capacity utilization, interacted with the $Futures_{product}$ indicator (Source: U.S. Geological Survey). Column 4 adds trade controls: annual growth rates of steel imports into the U.S. and global steel production, interacted with the $Futures_{product}$ indicator (Source: U.S. Geological Survey). Standard errors clustered by publication date are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

estimates remain significant at the 1% level throughout, and magnitudes are remarkably stable across specifications.

To assess the validity of the empirical design and study the dynamics of

(a) Table 2: Price dispersion DiD

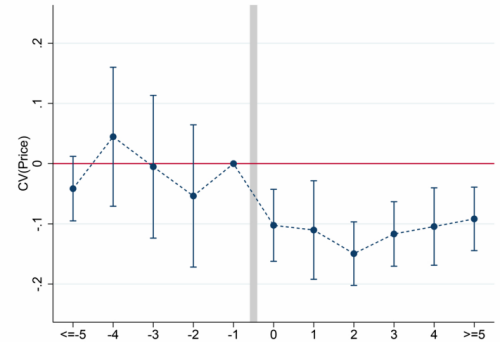


Figure 2
Physical Product Market Price Dispersion - Dynamic Effects
This figure shows estimates of the effect of centralized futures markets on price dispersion ($CV(Price)$) obtained from a difference-in-differences analysis around futures introductions. Specifically, the figure shows the estimated β_t coefficients and 95% confidence intervals from the following regression:

$$CV(Price)_{k,p,t} = \sum_{j=0}^5 (\beta_j Futures_{product,k,p}) \mathbb{1}\{t=\tau+j\} + \alpha_{k,p} + \alpha_{k,j} + \epsilon_{k,p,t}.$$

(b) Figure 2: Dynamic price dispersion

5.3 Table 3: Producer Commodity Hedging and Figure 3: Producer Commodity Hedging - Dynamic Effects

Read:

- Table 3 shows that treated firms become more likely to disclose commodity hedging.
- The coefficient on $Post \times Futures_{Firm}$ is about 0.215 to 0.258, implying a large increase in the probability of discussing commodity derivatives.
- The effect remains after baseline controls and interactions with year dummies.

- Figure 3 shows little evidence of differential pre-trends and a clear post-treatment increase.

Interpretation: The hedging result supports the risk-management channel. Since exchange-traded futures reduce counterparty risk and increase liquidity relative to OTC arrangements, treated producers appear to use commodity derivatives more after futures markets become available.

Economic message: Futures centralization does not only inform buyers. It also changes firms' financial toolkits, making it easier for producers to hedge exposure to steel prices.

Table 3
Producer commodity hedging: DiD estimation around futures introductions

	(1)	(2)	(3)
	Hedge (1/0)		
Post×Futures _{firm}	0.258*** (0.089)	0.248*** (0.091)	0.215** (0.096)
Firm FE	Yes	Yes	Yes
Year FE	Yes	No	No
Industry×Year FE	No	Yes	Yes
Controls×Year FE	No	No	Yes
R ²	.689	.695	.701
Observations	2,993	2,993	2,993

This table presents estimates of the effect of centralized futures markets on producers' commodity hedging activity (*Hedge (1/0)*) obtained from a difference-in-differences analysis around futures introductions. *Hedge (1/0)* is an indicator equal to one for firms discussing usage of commodity derivatives in their annual report. *Futures_{firm}* is an indicator equal to one for firms selling hot-rolled coil (HRC) around the 2008 introduction of HRC futures and for firms selling ferrous scrap around the 2012 introduction of busheling scrap futures (BUS). *Post* is an indicator equal to one in the years $y=0,1,\dots,4$, where $y=-1$ denotes the last year before the futures started trading. Controls are measured in the last quarter before the respective introduction and include the log of assets, firm age, and sales growth. Standard errors clustered by firm are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

the net effect of the centralized futures markets is that treated firms isolate their profits more from steel price risk in the context of the NYMEX steel futures.²⁶

4.3. Cost sensitivity of market shares

In the previous sections, I document that the futures markets decrease prices

(a) Table 3: Producer commodity hedging

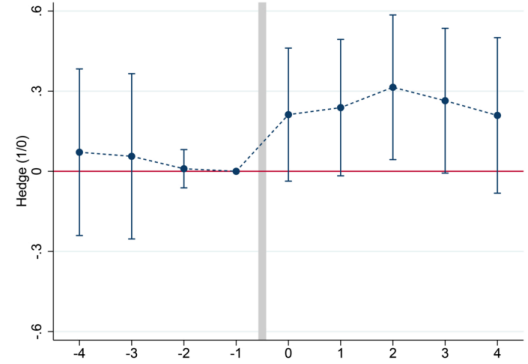


Figure 3
Producer Commodity Hedging - Dynamic Effects
This figure shows estimates of the effect of centralized futures markets on firms' likelihood to discuss commodity hedging in their annual report (*Hedge (1/0)*) obtained from a difference-in-differences analysis around futures introductions. Specifically, the figure shows the estimated β_t coefficients and 95% confidence intervals from the following regression:

(b) Figure 3: Dynamic hedging effects

5.4 Table 4: Product Market Market Share Allocation and Table 5: Physical Product Market Prices

Read:

- Table 4 tests whether market share becomes more sensitive to producer costs.
- The triple interaction $\text{Post} \times \text{FuturesFirm} \times \text{EAF} \times \text{Iron/Scrap}$ is positive and statistically significant, with coefficients around 0.054 to 0.078.
- This means EAF producers gain more market share after futures introduction when scrap becomes cheaper relative to iron ore.
- Table 5 shows that log product prices decline by about 3.2 to 4.0 percent for treated products after futures introductions.
- Product-price results remain robust to demand, supply, and trade controls.

Interpretation: Tables 4 and 5 connect transparency and hedging to competition. Table 4 shows that buyers and market shares respond more to cost differences after futures prices become available. Table 5 shows the aggregate consequence: treated product prices fall.

Economic message: Centralized futures markets appear to make the physical steel market more competitive. Low-cost producers capture more demand, and average seller markups decline.

Table 4
Market share sensitivity to cost: DID estimation around futures introductions

	(1)	(2)	(3)
	Market share		
Post×Futures _{firm} ×EAF×Iron/Scrap	0.054*** (0.016)	0.078*** (0.023)	0.073*** (0.020)
Firm FE	Yes	Yes	Yes
Year-quarter FE	Yes	No	No
Industry×Year-quarter FE	No	Yes	Yes
Controls×Year-quarter FE	No	No	Yes
Interactions	Yes	Yes	Yes
Sample	HRC	HRC	HRC
R ²	.980	.980	.981
Observations	1,419	1,419	1,419

This table presents estimates of the effect of centralized futures markets on the sensitivity of market share (*Market share*) to cost obtained from a difference-in-differences analysis around the hot-rolled coil (HRC) futures introduction. *Futures_{firm}* is an indicator equal to one for firms selling HRC around the 2008 introduction of HRC futures. *Post* is an indicator equal to one in the quarters $q=0, 1, \dots, 7$, where $q=-1$ denotes the last quarter before the futures started trading. *EAF* is an indicator equal to one for electric arc furnace (EAF) producers. EAF producers use steel scrap as their primary raw material whereas basic oxygen furnace (BOF) producers use primarily iron ore. *Iron/Scrap* is the quarterly ratio of iron ore to scrap prices. Controls are measured in $q=-1$ and include the log of assets, firm age, and sales growth. Standard errors clustered by firm are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

where i indexes firms, q indexes year-quarters, and j indexes industries at the 3-digit NAICS level. *Futures_{firm}* equals one for HRC producers. *Post* equals one starting in Q4 2008, and *EAF* equals one for producers using

(a) Table 4: Market share allocation

Table 5
Physical product market prices: DID estimation around futures introductions

	(1)	(2)	(3)	(4)	(5)
	Ln(Price)				
Post×Futures _{product}	-0.036*** (0.004)	-0.032*** (0.005)	-0.039*** (0.006)	-0.036*** (0.004)	-0.040*** (0.007)
Product FE	Yes	Yes	Yes	Yes	Yes
Publication Date FE	Yes	Yes	Yes	Yes	Yes
Demand Controls	No	Yes	No	No	Yes
Supply Controls	No	No	Yes	No	Yes
Trade Controls	No	No	No	Yes	Yes
R ²	.917	.917	.917	.917	.918
Observations	19,653	19,653	19,653	19,653	19,653

This table presents estimates of the effect of centralized futures markets on physical product market prices (*Ln(Price)*) obtained from a difference-in-differences analysis around futures introductions. *Futures_{product}* is an indicator equal to one for hot-rolled coils (HRC) around the 2008 introduction of HRC futures and for busheling scrap (BUS) around the 2012 introduction of BUS futures. Control products are cold-rolled coil, plate, heavy-melting scrap, and shredded scrap. Prices are collected and published by *SteelBenchmark* every 2 weeks. *Post* is an indicator equal to one for publication dates $t=0, 1, \dots, 40$, where $t=-1$ denotes the last publication date before the futures started trading. Column 2 adds controls to capture steel demand: quarterly GDP growth and output growth of key steel-consuming sectors (i.e., automotive, construction, machinery, appliances), interacted with the *Futures_{product}* indicator (Source: Bureau for Economic Analysis). Column 3 adds controls to capture steel supply: annual growth rates of U.S. steel production (in metric tons) and capacity utilization, interacted with the *Futures_{product}* indicator (Source: U.S. Geological Survey). Column 4 adds trade controls: annual growth rates of steel imports into the U.S. and global steel production, interacted with the *Futures_{product}* indicator (Source: U.S. Geological Survey). Standard errors clustered by publication date are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5 shows that product market prices decrease by 3% to 4% for treated

(b) Table 5: Product prices

5.5 Figure 4: Physical Product Market Prices - Dynamic Effects and Table 6: Operating Profits

Read:

- Figure 4 shows the dynamics of the log price effect.
- Product prices do not show strong differential movement before treatment, while post-treatment coefficients move negative.
- Table 6 shows that treated producers' operating profits fall by about 1.6 to 2.2 percentage points of assets.
- The profit result remains robust to controls for firm characteristics, industry-quarter fixed effects, input price exposure, import competition, and acquisitions.

Interpretation: Figure 4 supports the timing of the price effect, while Table 6 shows that lower prices are not merely offset by better operations. The profit decline indicates that increased competition reduces producer rents.

Economic message: The futures market may help firms hedge, but the competitive effect dominates for producers: transparency and risk management make the market tougher, not more profitable.

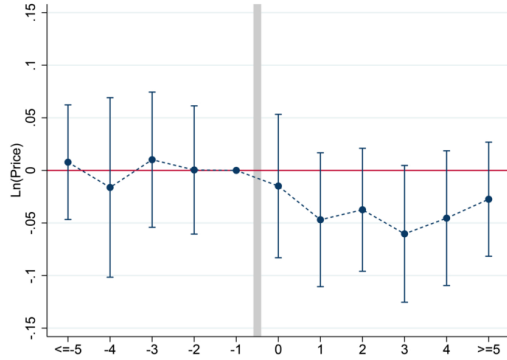


Figure 4
Physical Product Market Prices - Dynamic Effects
This figure shows estimates of the effect of centralized futures markets on physical product market price ($\ln(\text{Price})$) obtained from a difference-in-differences analysis around futures introductions. Specifically, the figure shows the estimated β_t coefficients and 95% confidence intervals from the following regression:

$$\ln(\text{Price})_{i,t} = \sum_{q=-5}^5 \beta_q \text{Future}_{i,t,q} + \text{Controls}_{i,t} + \text{Time Fixed Effects}_{i,t} + \text{Industry Fixed Effects}_{i,t} + \text{Error}_{i,t}$$

(a) Figure 4: Dynamic product prices

Table 6
Producer profits: DID estimation around futures introductions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Profit						
Post $\times$ Futures $_{firm}$	-0.016*** (0.004)	-0.019*** (0.004)	-0.021*** (0.005)	-0.022*** (0.004)	-0.019*** (0.005)	-0.019*** (0.005)	-0.019*** (0.005)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Business cycle controls	No	No	Yes	Yes	Yes	Yes	Yes
Iron/scrap price controls	No	No	No	Yes	Yes	Yes	Yes
Import controls	No	No	No	No	Yes	Yes	Yes
Segment controls	No	No	No	No	No	Yes	Yes
Excluding M&A	No	No	No	No	No	No	Yes
R^2	.602	.629	.652	.669	.670	.677	.684
Observations	5,095	5,095	4,953	4,942	4,942	4,942	4,591

This table presents estimates of the effect of centralized futures markets on producers' operating profits scaled by total assets (Profit) obtained from a difference-in-differences analysis around futures introductions:

$$\text{Profit}_{i,k,t,q} = \beta \cdot \text{Post} \cdot \text{Futures}_{firm,k,t} + \sum_{\tau=-7}^7 (\theta_{\tau}^q \cdot X_{k,t}) 1\{q=\tau\} + a_{k,t} + a_{k,j,q} + e_{k,t,q}$$

Futures introductions (HRC, BUS) are indexed by k , firms by i , year-quarters by q , and 3-digit NAICS industries by j . $\text{Profit}_{i,k,t,q}$ is the operating profit of firm i in year-quarter q scaled by beginning-of-quarter total assets. Futures_{firm} is an indicator equal to one for firms selling hot-rolled coil (HRC) around the 2008 introduction of HRC futures and for firms selling ferrous scrap around the 2012 introduction of busheling scrap futures (BUS). Post is an indicator equal to one in the quarters $q=0,1,\dots,7$, where $q=-1$ denotes the last quarter before the futures started trading. Column 2 adds the baseline controls (log of assets, firm age, and sales growth, measured in $q=-1$ and interacted with the time fixed effects). Column 3 adds business cycle controls (firms' sales beta with respect to aggregate GDP, automotive, construction, machinery, and appliance sector growth all interacted with the time fixed effects). Column 4 adds controls for iron ore and scrap prices (firms' profit and input cost beta with respect to iron ore relative to scrap prices and a dummy for EAF producers, all interacted with the time fixed effects, as well as interactions of the Futures_{firm} dummy with the quarterly ratio of iron ore to scrap prices and with the log of quarterly iron ore prices); column 5 adds import competition controls (the Futures_{firm} dummy interacted with industry-level import competition from all countries and from China, India, and Japan); and column 6 adds segment controls (the share of foreign sales and the share of sales in the main industry interacted with time fixed effects). Column 7 excludes observations involved in M&A activity. Standard errors clustered by firm are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 6: Operating profits

5.6 Figure 5: Operating Profits - Dynamic Effects, Table 7: Stock Market Valuation, and Figure 6: Event Study Around News Related to Futures Introductions

Read:

- Figure 5 shows that treated and control firms have similar profit trends before futures introduction, followed by lower profitability for treated firms.
- Table 7 shows that firms selling treated products experience negative abnormal returns around futures-related news.
- The CAR coefficients are about -3.7% to -4.9%, depending on the benchmark model.
- Figure 6 shows the valuation effect dynamically around event dates, with negative post-event CARs for treated firms.

Interpretation: The valuation evidence is a forward-looking test. Investors appear to anticipate that futures introductions will reduce future producer rents. Because the result is measured around news events, it is less vulnerable to slow-moving confounders than long-panel profit regressions.

Economic message: Product market competition is capitalized into equity values. The stock market views futures introductions as bad news for affected producers, consistent with lower future margins.

regarding whether and when futures contracts would be introduced, and which products would be covered.

I consider five significant events for the 2008 HRC futures introduction. The first is a contract signed between NYMEX and World Steel Dynamics to use *SteelBenchmark* prices for settling HRC futures. However, this contract did not materialize because NYMEX was acquired by CME Group before implementation.³⁰ Prior to the merger, CME had instructed the CRU Group, a commodity research firm, to develop a price index to settle HRC futures. Following the merger, NYMEX's initial plans were abandoned, and the new formed CME/NYMEX entity announced the launch of HRC futures in February 2008, settling against the price index developed by CRU. For the 2012 BUS futures introduction, key events include the signing of a licensing agreement between CME/NYMEX and American Metal Market (AMM) for the AM ferrous scrap index, and the official announcement that U.S. Midwest busheling scrap futures would be traded on the exchange.³¹

I estimate cumulative abnormal returns (CAR) for treated and control producers around these events using the market model, the capital asset pricing model (CAPM), the Fama-French three-factor model (Fama and French 1999), and the Carhart four-factor model (Carhart 1997). I then estimate the following OLS regression:

(a) Figure 5: Dynamic profits

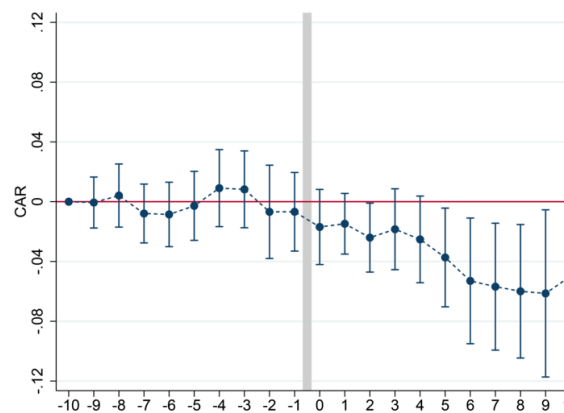


Figure 6
Event Study Around News Related to Futures Introductions
This figure shows estimates of the effect of centralized futures markets on firms' stock market valuations (CAR) obtained from a difference-in-differences analysis around news increasing the likelihood of futures contracts. Specifically, the figure shows the estimated β_T coefficients and 95% confidence intervals from the following regression:

Table 7
Event study around news related to futures introductions

	(1)	(2)	(3)	(4)
	CAR _{-2,+2}			
Futures _{firm}	-0.049*** (0.010)	-0.037*** (0.012)	-0.039*** (0.015)	-0.039*** (0.013)
Firm FE	Yes	Yes	Yes	Yes
Event date FE	Yes	Yes	Yes	Yes
Model	Market-adj.	CAPM	3-factor	4-factor
R ²	.257	.241	.225	.233
Observations	1,106	1,106	1,106	1,106

This table presents OLS estimates of regressing producers' cumulative abnormal returns (CAR) during the 5-day window around news increasing the likelihood of hot-rolled coil (HRC) and busheling scrap (BUS) futures on an indicator variable for affected firms: *Futures_{firm}* is an indicator equal to one for firms selling hot-rolled coil (HRC) on events relating to the 2008 introduction of HRC futures and for firms selling ferrous scrap on events relating to the 2012 introduction of busheling scrap futures (BUS). Standard errors clustered by firm are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Figure 6 presents the dynamics of the effect during the 21-day window surrounding the events. While treated and control firms exhibit similar abnormal returns prior to the event date, treated firms' abnormal returns begin to decrease relative to control firms around the news release. The effect appears to be fully priced in 6 days after the event. Taken together, the results are

(b) Table 7: Stock market valuation

5.7 Table 8: Separate DiD Estimation around HRC and BUS Futures Introductions and Table 9: Placebo Tests for Non-U.S. Producers

Read:

- Table 8 estimates effects separately for HRC and BUS introductions.
- HRC results show lower price dispersion, higher hedging, lower prices, and lower profits.
- BUS results show the same signs and are statistically significant for price dispersion, hedging, prices, and profits.
- The BUS-only estimates are especially important because they avoid the 2008 financial crisis and commodity boom-bust window.
- Table 9 uses non-U.S. producers as a placebo group.
- Non-U.S. treated-product producers do not show significant changes in hedging, market share, profits, or CARs around the U.S. futures introductions.

Interpretation: Table 8 reduces concerns that the main estimates are only driven by the HRC introduction during the Great Recession. Table 9 addresses the alternative explanation that treated products were hit by global shocks unrelated to NYMEX futures. Together, they strengthen the causal interpretation.

Economic message: The effects are tied to the U.S. futures market rather than to generic changes in global steel conditions.

Table 8
Separate DiD estimation around HRC and BUS futures introductions

	(1)	(2)	(3)	(4)	(5)
<i>A. HRC</i>	SD(Price)	CV(Price)	Hedge (1/0)	Ln(Price)	Profit
Post×Futures _{product}	-55.875*** (13.249)	-0.049*** (0.015)		-0.034*** (0.005)	
Post×Futures _{firm}			0.206* (0.120)		-0.023** (0.009)
R ²	.841	.894	.652	.900	.648
Observations	222	222	1,551	8,856	2,508
<i>B. BUS</i>	SD(Price)	CV(Price)	Hedge (1/0)	Ln(Price)	Profit
Post×Futures _{product}	-26.081*** (6.241)	-0.064*** (0.012)		-0.038*** (0.008)	
Post×Futures _{firm}			0.301** (0.127)		-0.018*** (0.006)
R ²	.523	.440	.721	.921	.720
Observations	486	486	1,442	10,797	2,434
Product FE	Yes	Yes	No	Yes	No
Publication date FE	Yes	Yes	No	Yes	No
Firm FE	No	No	Yes	No	Yes
Year FE	No	No	Yes	No	No
Industry×Year-quarter FE	No	No	No	No	Yes
Controls×Year-quarter	No	No	No	No	Yes

This table presents the main results separately for the hot-rolled coil (panel A) and busheling scrap (panel B) futures introductions. *Futures_{product}* is an indicator equal to one for hot-rolled coils (HRC) around the 2008 introduction of HRC futures and for busheling scrap (BUS) around the 2012 introduction of BUS futures. *Futures_{firm}* is an indicator equal to one for firms selling HRC around the 2008 introduction of HRC futures and for firms selling ferrous scrap around the 2012 introduction of BUS futures. *Post* is an indicator equal to one after futures started trading. Standard errors clustered by publication date in columns 1, 2, and 4 and by firm in columns 3 and 5 are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 8: HRC and BUS separately

Table 9
Placebo tests: DiD estimation around futures introductions for non-U.S. producers

	(1)	(2)	(3)	(4)
	Hedge (1/0)	Market Share	Profit	CAR _{-2,+2}
Post×Futures _{firm}	0.009 (0.067)		-0.000 (0.003)	
Post×Futures _{firm} ×EAF×Iron/Scrap		0.008 (0.023)		
Futures _{firm}				-0.004 (0.022)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No
Year-quarter FE	No	Yes	Yes	No
Interactions	No	Yes	No	No
Event date FE	No	No	No	Yes
R ²	.719	.943	.661	.316
Observations	4,054	2,394	7,303	1,804

This table presents placebo tests in a sample of non-U.S. firms. Columns 1 to 4 estimate specifications akin to column 1 of Tables 3, 4, 6, and 7 in a sample of non-U.S. producers, operating in countries without steel futures introduction during the sample period (excluding China, India, South Korea, Turkey, the UAE, and the Black Sea region). *Futures_{firm}* is an indicator equal to one for firms selling HRC around the 2008 introduction of HRC futures and for firms selling ferrous scrap around the 2012 introduction of BUS futures. *Post* is an indicator equal to one after the HRC and BUS futures started trading in the United States. *EAF* is an indicator equal to one for electric arc furnace (EAF) producers. EAF producers use steel scrap as their primary raw material whereas basic oxygen furnace (BOF) producers use primarily iron ore. *Iron/Scrap* is the quarterly ratio of iron ore to scrap prices. The sample in column 2 is restricted to the HRC introduction. Standard errors clustered by firm are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

sample. Additionally, treated firms' stock prices do not react differently to

(b) Table 9: Non-U.S. placebo tests

6 Interpretive synthesis

6.1 Why the paper is about real effects, not only financial market quality

The paper starts from a financial-market innovation: moving derivatives trading from decentralized or absent markets to a centralized exchange. The key contribution is showing that this innovation affects the real product market. Price dispersion, hedging, market shares, product prices, operating profits, and stock valuations all change in a coherent direction.

6.2 How the two channels fit together

The reference price channel explains why buyers bargain better and prices converge. The risk-management channel explains why sellers can use futures to stabilize cash flows and compete more aggressively. These two channels both point toward stronger product-market competition, even though one operates through buyer information and the other through producer financing and risk management.

6.3 Why producer losses do not imply social losses

Lower producer profits and valuations indicate reduced rents for sellers, not necessarily lower welfare. Buyers may benefit from lower prices and less dispersion; low-cost producers gain market share; and resources may be allocated more efficiently. The paper does not fully compute welfare, but the pattern is consistent with a transfer away from producer rents and toward buyers and efficient producers.

6.4 External validity

The setting is most relevant for markets with bilateral negotiation, limited price transparency, feasible futures contracts, and meaningful price risk. The paper is less likely to apply to spot markets with already transparent prices, low customer search frictions, or products without viable centralized derivatives.

7 Short summary

- The paper studies the real effects of centralized derivative markets using the introduction of HRC and BUS steel futures.
- The empirical design compares treated steel products with related control products around futures introductions.
- Price dispersion falls after futures markets arrive, supporting the reference price channel.
- Treated producers become more likely to hedge with commodity derivatives, supporting the risk-management channel.
- Market share becomes more sensitive to relative production costs, showing that low-cost producers gain share.
- Product prices fall by about 3–4 percent.
- Operating profits fall by about 1.6–2.2 percentage points of assets for treated producers.
- Stock prices of affected producers fall by about 4–5 percent around news that increases the likelihood of futures contracts.
- Separate HRC and BUS results show similar patterns.
- Non-U.S. placebo tests show no comparable effects, mitigating concerns about global steel shocks.
- The overall evidence indicates that centralized futures markets foster competition in the underlying product market.

8 Conclusion

8.1 Core conclusion

Martin shows that centralized derivative markets can have important real effects on the underlying product markets. Steel futures reduce price dispersion, increase producer hedging, reallocate market share toward low-cost producers, lower product prices, reduce operating profits, and depress the stock valuations of affected producers.

8.2 Economic intuition

The intuition is that futures prices become public reference prices in an otherwise opaque bilateral bargaining market. Buyers use those reference prices to negotiate more effectively and identify better deals. At the same time, producers gain access to improved hedging tools. Together, these forces increase competition, which lowers prices and producer rents.

8.3 What the paper does not claim

The paper does not claim that all centralized derivatives always reduce producer value, or that futures markets are socially harmful. The results are specific to settings where futures markets create useful price benchmarks and where product markets are opaque enough for transparency to matter.

8.4 Final takeaway

The key lesson is that financial market design can reshape real product markets. Centralized futures markets are not just venues for financial trading; they can change bargaining, risk management, market shares, prices, profits, and valuations in the industries whose products they reference.